and IDE is on its way to a speedy exit. One of the things helping the SATA cause is the fact that even entry-level motherboards ship with SATA controllers today. Granted, IDE drives are still being sold, but the fact remains that there are very few of them, and future development prospects are too dim to warrant a shootout.

Of the 21 SATA drives we pitted against each other, two were from Hitachi, two from Maxtor, four from Samsung, three from Seagate, and there was a large contingent of 10 from Western Digital. All the brands were thus represented adequately to give you as wide a perspective as we possibly can about the drives available in the market today.

Features

Capacity

The Hitachi Deskstar 7K1000 with a huge 1 TB capacity was the largest internal SATA drive we've ever laid our hands on. This drive, featuring perpendicular recording technology, featured a platter density of 250 MB—the highest we have seen so far, shattering the previous record of 188 set by the Seagate Barracuda 7200.9 ST3750640AS.

SATA I or SATA II?

While most of the drives were SATA II compliant, a few of the older drives, such as the Maxtor DiamondMax 10 250 GB, Seagate Barracuda 7200.8 ST3400832AS, and the two Western Digital Raptors were SATA I. Naturally, you would expect these to lag behind in the performance tests, but this was not entirely the case; the Raptors are blistering fast even when compared with the latest SATA II drives.

Though the SATA standard is forward-as well as backward-compatible, there are still some known issues plaguing certain older controllers. Some manufacturers such as Seagate, Samsung, and Western Digital remedy this situation by providing pins that can be jumpered to set the mode of the drive, while some such as Hitachi follow a different method. To set a Hitachi drive to SATA/150 mode, you first need to go to their Web site and download the Feature Tool, which is a bootable ISO image containing the tool. Once created and booted through, it guides you through the process. This is obviously tedious compared to the jumper approach.